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Determination, correlation, and mechanistic interpretation of effects of hydrogen addition on laminar flame speeds of hydrocarbon–air mixtures

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Abstract

The stretch-affected propagation speeds of expanding spherical flames of *n*-butane–air mixtures with hydrogen addition were measured at atmospheric pressure and subsequently processed through a nonlinear regression analysis to yield the stretch-free laminar flame speeds. Based on a hydrogen addition parameter (R_H) and an effective fuel equivalence ratio (ϕ_F), these laminar flame speeds were found to increase almost linearly with R_H, for ϕ_F between 0.6 and 1.4 and R_H from 0 to 0.5, with the slope of the variation assuming a minimum around stoichiometry. These experimental results also agree well with computed values using a detailed reaction mechanism. Furthermore, a mechanistic investigation aided by sensitivity analysis identified that kinetic effects through the global activation energy, followed by thermal effects through the adiabatic flame temperature, have the most influence on the increase in the flame speeds and the associated linear variation with R_H due to hydrogen addition. Nonequidiffusion effects due to the high mobility of hydrogen, through the global Lewis number, have the least influence. Further calculations for methane, ethene, and propane as the fuel showed similar behavior, leading to possible generalization of the phenomena and correlation.
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Keywords: Laminar flame speed; Hydrogen addition

1. Introduction

Because of its strong reactivity, hydrogen is an attractive additive in enhancing the combustion intensity of fuel–air mixtures. In particular, by extending the extinction and flammability limits of lean mixtures, hydrogen addition holds the potential to promote combustion efficiency and reduce pollutant formation; the former further facilitates its use as a fuel source in response to issues related to energy sustainability and climate change.

The fundamental combustion parameter that compactly characterizes and quantifies the effects of hydrogen addition is the laminar flame speed of the resulting mixture. In particular, Milton and Keck [1] first measured the laminar flame speeds of stoichiometric mixtures of hydrogen-enriched

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methane, acetylene, and propane with air using the constant-volume, expanding spherical flame method, with their results further discussed by Refael and Sher [2]. Recently measurements were also conducted for hydrogen-enriched natural gas [3] and propane [4], again using the expanding flame method. Results showed a weak dependence on hydrogen addition up to about 50% hydrogen mole fraction, and a rapid increase to approach the hydrogen–air flame speed beyond 80% addition. Yu et al. [5] used the symmetric counterflow flame to measure the laminar flame speeds for methane–air and propane–air mixtures with hydrogen addition, and found that, regardless of whether the mixture was lean or rich, the increased flame speeds can be approximately linearly correlated with a hydrogen addition parameter R_H, which will be defined later This work has enjoyed substantial follow-on investigation, ostensibly because of the interest in hydrogen combustion and the simplicity of the correlation.

There are several motivations for the present study. First, in view of the substantial progress in experimental and computational combustion research since the publication of Ref. [5] in 1986, and the interest this work has generated in research on hydrogen combustion and utilization, it behooves us to re-examine and extend the results and concepts advanced therein. In particular, the accuracy associated with the determination of laminar flame speeds from stretched flames has further improved since the mid-1980s, including the use of nonlinear extrapolation techniques to accurately remove stretch effects from data acquired from expanding flames [6]. Furthermore, while the investigation of Ref. [5] was strictly experimental in nature, computational simulation of the 1D freely-propagating laminar flame, with the concomitant determination of the laminar flame speed, is now not only a routine activity, but is actually a useful tool in studies of this nature. It is therefore only compelling that the phenomena of interest be synergistically studied with inputs from computation.

Second, there has not been a systematic investigation and discourse on the cause(s) for the observed enhancement effect with hydrogen addition, apart from the obvious implication of the strong reactivity of hydrogen. In this regard we note that, while enhanced reactivity is certainly expected to be important, there are two additional factors whose significance also needs to be assessed. The first is thermal effect, caused by the inherently higher adiabatic flame temperatures of hydrogen–air mixtures as compared to those of the hydrocarbon–air mixtures, for the same equivalence ratio. The increased adiabatic flame temperature obviously would lead to a corresponding increase in the mixture reactivity, even assuming the same underlying reaction mechanism. The second is the high mobility of the hydrogen molecule compared to those of the hydrocarbons and air, causing diffusional modification of the mixture concentration in the flame structure. The net effect due to diffusional modification therefore also needs to be examined. Collectively, then, a comprehensive understanding of the factors enhancing flame propagation with hydrogen addition must consider the relative importance of the kinetic, thermal, and transport effects.

Finally, an intriguing issue that needs to be addressed is the approximate linearity of the correlation between the experimentally determined laminar flame speed and the addition parameter R_H identified in Ref. [5], especially considering that the amount of addition is not always very "small", and that the Arrhenius kinetics is intrinsically nonlinear.

In terms of the present experimental investigation, we have selected *n*-butane as the target fuel for this study because it is the largest gaseous fuel with opposite nonequidiffusion behavior as that of hydrogen. Furthermore, a reasonably comprehensive reaction mechanism has recently been developed for *n*-butane [7], allowing meaningful computational simulation, comparison with experimental data, and identification of the various factors influencing the phenomena of interest.

The experimental specifications are discussed in the next section, which is followed by presentation of the experimental results on *n*-butane. These results will be analyzed for linear correlation in Section 4, together with a detailed analysis of the sensitivity of the kinetic, thermal, and diffusion factors in response to hydrogen addition on the flame propagation. The computational study will then be extended to methane, ethene, and propane as the fuel, yielding interesting and useful generalization of this phenomenon.

2. Experimental specifications

We have adopted the spark-ignited, outwardly-propagating spherical flame in an enclosed chamber to acquire the stretched flame speed, from which the unstretched laminar flame speed is abstracted by eliminating stretch effects. Parallel experiments were conducted in two different apparatuses which were located at Xi'an Jiaotung University and at Princeton University, which will be, respectively, designated as Chamber J and Chamber P. All the experiments with and without hydrogen addition were conducted in Chamber J, while additional experiments on *n*-butane, with and without hydrogen addition, were also conducted in Chamber P to provide validation checks. Chamber P is a *dual-chamber*, *constant-pressure* apparatus that in recent years has been used to generate extensive amount of data on the laminar flame speeds of a variety of fuels, and its design is well documented [6,8]. Chamber

J is a *single-chamber*, *constant-volume* apparatus, and its detailed design is given in Ref. [4]. Briefly, it is a cylindrical chamber (180 mm diameter and 210 mm length) which can withstand an initial pressure of up to 10 atmospheres and can also be heated up to 500 K. The spherically-propagating flame was imaged using Schlieren photography and recorded by a high-speed digital camera operating at 2000 to 10,000 frames per second, depending on the flame speed. The spatial resolution of the image corresponds to about 0.1 mm. The hydrogen and *n*-butane used in the study were, respectively, of 99.995% and 99.95% purity.

To quantify the extent of hydrogen addition, we adopt the concentration parameters of Ref. [5]. Specifically, it is noted that for small to moderate amount of hydrogen addition in terms of oxygen consumption, there should always be enough air to facilitate the complete oxidation of hydrogen. Therefore in order to oxidize C_H amount of hydrogen, $C_H/(C_H/C_A)_{\text{stoic}}$ amount of air is needed, where C_H, C_F, and C_A are, respectively, the mole fractions of hydrogen, the hydrocarbon fuel, and air, such that $C_F + C_H + C_A = 1$, and where $(C_H/C_A)_{\text{stoic}} = 0.420$ is the stoichiometric hydrogen-to-air molar ratio. Consequently, if the remaining air is used to oxidize the fuel, an effective fuel–air equivalence ratio can be defined as

$$\phi_F = \frac{C_F/[C_A - C_H/(C_H/C_A)_{stoic}]}{(C_F/C_{A)_{stoic}}} \tag{1}$$

where $(C_F/C_A)_{\text{stoic}}$ is the stoichiometric fuel-to-air molar ratio, which is 0.032 for butane–air mixtures.

Based on the above reasoning, a parameter representing the relative amount of hydrogen addition can be defined as

$$R_H = \frac{C_H + C_H/(C_H/C_A)_{stoic}}{C_F + [C_A - C_H/(C_H/C_A)_{stoic}]} \tag{2}$$

which is the ratio of the amount of hydrogen plus the amount of air needed to totally oxidize it, to the amount of the hydrocarbon plus the amount of air available for its oxidation.

It is noted that while these two parameters are intuitively defined, we shall demonstrate in due course that they do facilitate data reduction and correlation, and as such are empirically viable parameters to measure and represent the effects of hydrogen addition.

Figure 1a relates this hydrogen addition parameter R_H to the molar fraction of hydrogen in the total fuel, namely $C_H/(C_H + C_F)$, for R_H up to 0.5 which is the maximum value considered in this study. It is seen that the amount of hydrogen addition in terms of molar fraction is quite substantial, up to 70–90% of the total fuel. On the other hand, since hydrogen is a much smaller molecule than butane, its heat of combustion per mole is correspondingly much smaller (57.8 vs. 635.2 kcal/mol). As such, a more indicative assessment of the amount of hydrogen addition is the molar ratio of oxygen needed to consume the hydrogen to the amount needed to consume both hydrogen and butane, as shown in Fig. 1b. It is then seen that based on this indicator hydrogen can be considered to be the minor component of the two fuels, around 0.27 even for $R_H = 0.5$.

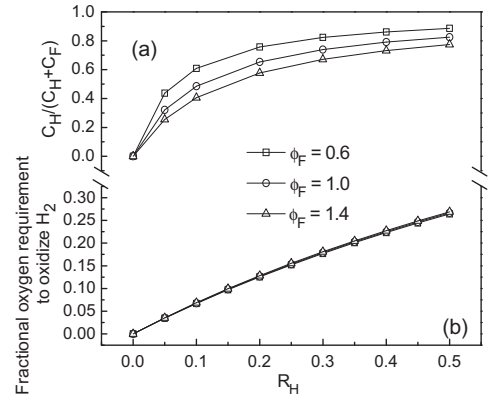

Fig. 1. Relations between the hydrogen addition parameter, R_H, and (a) hydrogen mole fraction in total fuel, and (b) fractional oxygen requirement to oxidize the hydrogen component.

3. Experimental results

3.1. Data analysis

The raw data obtained for a given run is the flame radius history, $r_f(t)$. Assuming the burned gas is motionless, the stretched flame speed and stretch rate were evaluated, respectively, through $s_b = dr_f/dt$ and $\kappa = (2/r_f)(dr_f/\text{dt}) = 2s_b/r_f$, where the subscript "*b*" designates the burned state. There are two major considerations in processing the recorded data so that the results determined are meaningful. First, the data during the initial period are affected by the ignition energy and transient, with the extent depending on the mixture Lewis number, *Le* [9,10]. Consequently, for lean and rich butane mixtures, whose Lewis numbers are, respectively, greater and smaller than unity, the minimum radius beyond which the ignition effect is dissipated was, respectively, chosen to be 10 mm and 6–9 mm in this work, depending on the flame speed trajectory, as suggested in Ref. [6]. Furthermore, there is also an upper limit in the flame radius for meaningful data analysis because of chamber confinement and asymmetry effects [11]. In this work, the pressure increase at the maximum radius is less than 5%, and the upper radius limit is chosen to be less than 25 mm, which is less than 30% of the wall radius and thus the

effect of chamber confinement can be neglected, as indicated in Ref. [11].

The second concern is the accuracy associated with the extraction of the laminar flame speed by eliminating stretch effects from the experimental data. While the conventional approach has been the linear extrapolation of the experimental stretched flame speed to zero stretch, it was demonstrated in Ref. [6] that for the expanding spherical flames a nonlinear extrapolation using the relation

$$\left(\frac{s_b}{s_b^o}\right)^2 \ln\left(\frac{s_b}{s_b^o}\right)^2 = -2\frac{L_b\kappa}{s_b^o} \tag{3}$$

is required for accuracy, where s_b^o is the downstream laminar flame speed and L_b the Markstein length. Thus by appropriately extrapolating s_b to $\kappa = 0$, s_b^0 was obtained from which the upstream laminar flame speed s_u^0 was correspondingly determined through $s_u^0 = \rho_b^0/\rho_u s_b^0$. For enhanced accuracy in fitting, an integrated form of Eq. (3),

$$t = A\left[E_1(\ln\xi^2) - \frac{1}{\xi^2\ln\xi}\right] + C \tag{4}$$

was actually used in the present data processing [6], where

$$A = \frac{2L_b}{s_b^o}, \quad r_f = -\frac{2L_b}{\xi\ln\xi}, \quad E_1(x) = \int_x^\infty \frac{e^{-z}}{z}dz. \tag{5}$$

3.2. Data validation

To validate the present measurement system and data reduction method, the laminar flame speed of butane–air mixtures at atmospheric pressure were first measured and compared with literature values as well as the numerical predictions. Figure 2 shows the present data obtained from both Chamber J and Chamber P, together with the experimental results of Bosschaart et al. [12] using the heat flux method and of Davis and Law [13] using the counterflow method but with linear extrapolation, and the numerical prediction of USC Mech II [7]. Results show close comparison among these results from different sources, with the data from Chamber J being slightly higher than those from Chamber P on the lean side, by 2–3 cm/s, and the data of Ref. [13] being slightly higher overall, ostensibly due to the use of linear extrapolation. Furthermore, USC Mech II seems to predict slightly higher values of laminar flame speeds than all the experimental results for lean mixtures. This level of agreement with slight systematic deviations from different sources is typical of the state of accuracy in the determination of laminar flame speeds.

3.3. Results on butane–air mixtures

Figure 3 shows the measured and calculated laminar flame speeds of butane–air mixtures with hydrogen addition, for both lean and rich cases. It is seen that the calculated and measured laminar flame speeds agree quite well with each. Furthermore, for ϕ_F up to about 1.4, they also increase almost linearly with increasing R_H. This linear result therefore corroborates that of Yu et al. for hydrogen addition in methane–air and propane–air mixtures.

Following Ref. [5], we then correlated the measured laminar flame speed with R_H by a linear equation

$$s_u^o(\phi_F, R_H) = s_u^o(\phi_F, 0) + k(\phi_F)R_H \tag{6}$$

such that the coefficient $k(\phi_F)$ represents the sensitivity of hydrogen addition on the flame speed. Yu et al. showed that $k(\phi_F)$ is largely independent of the effective fuel equivalence ratio ϕ_F. In view of the increased accuracy with the present experimentation and the extraction procedure, and the availability of computational results, we have re-examined this issue and found that k actually

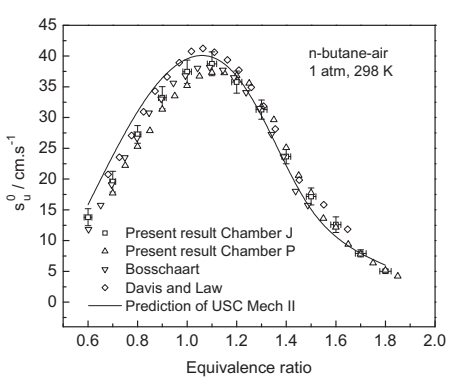

Fig. 2. Experimental and calculated laminar flame speeds of butane–air mixtures in the standard atmosphere.

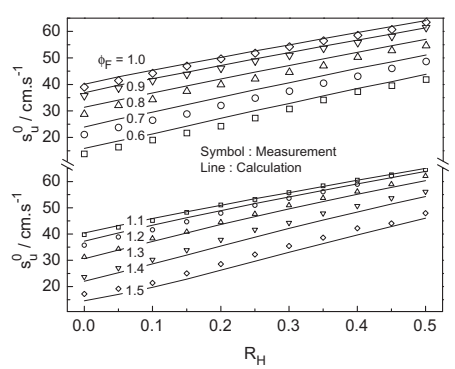

Fig. 3. Measured and calculated laminar flame speeds of mixtures of butane–air with hydrogen addition, as function of R_H.

depends on ϕ_F, exhibiting a minimum around $\phi_F = 1$, as shown in Fig. 4. Such a dependence is reasonable by recognizing that the positive influence of hydrogen addition should be stronger for off-stoichiometric, weaker burning situations, which in turn implies the minimal influence around stoichiometry.

4. Mechanistic interpretation of linearity

The existence of a linear variation between s_u^o and R_H not only is of practical utility, but is also fundamentally intriguing. In particular, since the amount of hydrogen addition is actually not very small, as shown in Fig. 1, and because of the sensitivity of Arrhenius kinetics, it is not clear that the linear variation can be simply explained as the Taylor expansion of a small parameter. Furthermore, since the potential influence of hydrogen addition can be considered on the basis of thermal, kinetic, and diffusion effects, as mentioned earlier, it is of interest to identify their relative importance.

To assess their relative importance, we recognize that the laminar burning flux, which is the density-weighted flame speed and is the eigenvalue for flame propagation, is given by:

$$f^2 \sim Le\ \exp(-E_a/R^0 T_{ad}) \tag{7}$$

in which the Lewis number Le, the activation energy E_a, and the adiabatic flame temperature T_{ad} can be considered to, respectively, represent the diffusion, kinetic, and thermal effects.

4.1. Determination of T_{ad}, E_a, and Le

The adiabatic flame temperature as a function of R_H is shown in Fig. 5. It is seen that it increases monotonically with increasing R_H, with a variation that is slightly weaker than linear. Furthermore, this dependence is weakened as the mixture is close to stoichiometry.

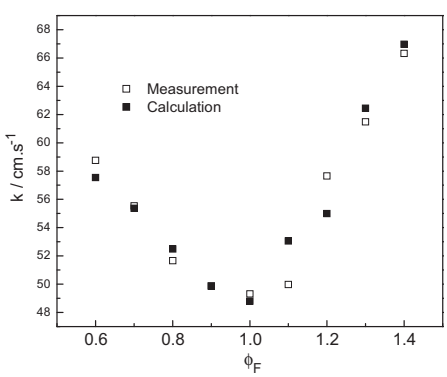

Fig. 4. Linearity coefficient, k, quantifying the increase in flame speed with R_H.

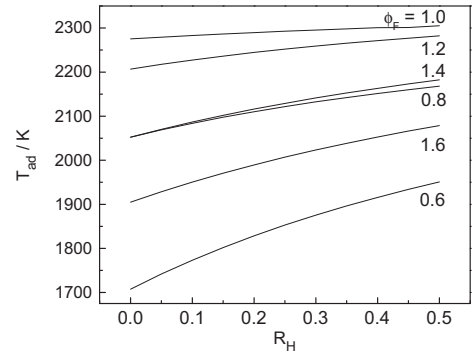

Fig. 5. Calculated adiabatic flame temperatures as function of R_H.

The global activation energy, E_a, can be extracted from the laminar burning flux, f, through $E_a = -2R^0 d\ln(f)/d(1/T_{ad})$ for sufficiently off-stoichiometric mixtures [14,15], where R^0 is the universal gas constant. This can be accomplished numerically by slightly perturbing the inert composition. For near-stoichiometric mixtures the E_a is interpolated from the off-stoichiometric mixtures [16]. Figure 6 shows the extracted E_a as a function of R_H for different ϕ_F. It is seen that for all fuel equivalence ratios, the global activation energy decreases with increasing hydrogen addition, indicating that the global reaction is promoted, which is reasonable. The variation is weaker for near-stoichiometric mixtures.

The Lewis number for the entire fuel mixtures, Le_{HF}, was evaluated from

$$Le_{HF} = 1 + (q_H + q_F)^{-1}[(Le_H - 1)q_H + (Le_F - 1)q_F]$$

which was derived in Ref. [17], where Le_H and Le_F are the Lewis numbers of hydrogen and butane, and q_H and q_F are the corresponding heat releases.

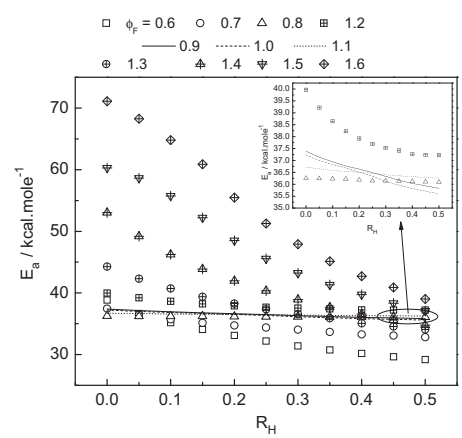

Fig. 6. Extracted global activation energies as function of R_H; inset shows a magnified view for near-stoichiometric mixtures.

The overall effective Lewis number, Le_{Eff}, was evaluated through the weighted average of Le_{HF} and Le_O, $Le_{Eff} = 1 + [(Le_E - 1) + A(Le_D - 1)]/(1 + A)$, given by Bechtold and Matalon [18], where $A = Ze|\phi - 1|$ is a parameter representing the stoichiometry, Ze the Zeldovich number, ϕ the equivalent ratio of the mixture, and Le_E and Le_D, respectively, represents the Lewis number of the excess and deficient reactant in terms of the oxidizer and the total fuel. Figure 7 shows the effective Lewis number as a function of R_H. It is seen that for lean and rich cases, hydrogen addition, respectively, decreases and increases the Lewis number, which is reasonable because of the very small and large Lewis numbers for lean and rich hydrogen–air mixtures, respectively.

4.2. Sensitivity of laminar burning flux

Having determined the dependence of Le, E_a, and T_{ad} on R_H, the influence of hydrogen addition on the laminar flame speed, or rather the laminar burning flux which is the eigenvalue for flame propagation, due to diffusion, kinetic, and thermal effects can be evaluated through sensitivity analysis. Thus using Eq. (4), the overall sensitivity coefficient is given by,

$$\frac{\partial \ln f}{\partial \ln R_H} \sim \frac{1}{2Le}\frac{dLe}{d\ln R_H} - \frac{1}{2T_{ad}}\frac{dT_a}{d\ln R_H} + \frac{T_a}{2T_{ad}^2} \times \frac{dT_{ad}}{d\ln R_H} \quad (8)$$

where $T_a = E_a/R^o$ is the activation temperature and the three terms on the right hand side, respectively, represent the individual sensitivity coefficients for diffusion, kinetic, and thermal effects. Figure 8 shows the comparison of these sensitivity coefficients as functions of R_H for three representative lean, stoichiometric and rich cases. It is seen that for all ϕ_F, the kinetic term is the largest sensitivity factor in increasing the burning flux. The sensitivity factor for the thermal effect is smaller but nevertheless still significant, while that for the diffusion effect is substantially smaller and can also assume negative values for lean mixtures. Furthermore, all the three sensitivity factors at the stoichiometric condition assume values one order smaller than those at the off-stoichiometric conditions of $\phi_F = 0.6$ and 1.4. This corroborates the earlier result that the laminar flame speed shows the weakest dependence on R_H at near-stoichiometric condition, shown in Fig. 4.

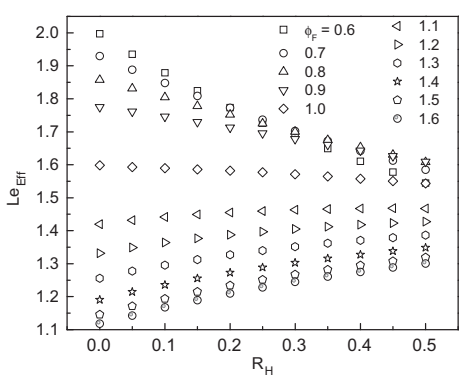

Fig. 7. Calculated effective Lewis number as function of R_H.

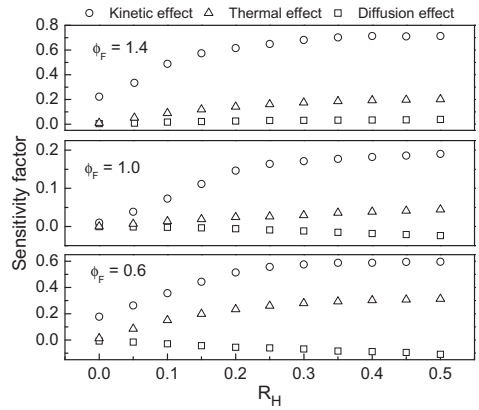

Fig. 8. Sensitivity coefficients measuring kinetic, thermal, and diffusion effects on the laminar flame speeds, with hydrogen addition, as functions of R_H.

4.3. Demonstration of linearity

Knowing the variations of Le, E_a, and T_{ad} with R_H, we can now evaluate the laminar burning flux, f, given by Eq. (5), as a function of R_H. Figure 9 shows that f indeed increases nearly linearly with R_H for the three fuel equivalence ratios of 0.6, 1.0, and 1.4, and that, the coefficient of the linear term is smaller for the stoichiometric

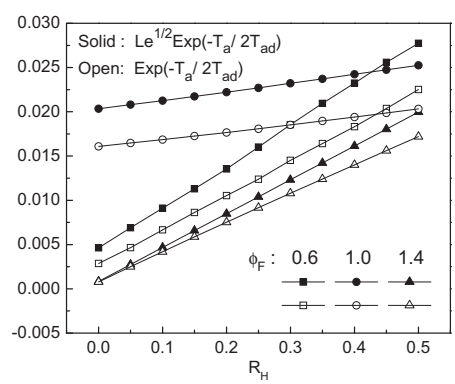

Fig. 9. Demonstration of the influence of the kinetic, thermal, and diffusion effects on the linearity of correlation, respectively, through the activation temperature, adiabatic flame temperature, and Lewis number.

mixture than those of off-stoichiometric mixtures, in agreement with our earlier observations. Furthermore, since the sensitivity to the *Le* effect is weak, the same extent of linearity is observed in Fig. 9 by suppressing the *Le* variation. This therefore demonstrates that the linear variation is primarily a consequence of combined kinetic and thermal effects.

5. Extension to other fuels

Additional computation has also been performed for methane, ethene, and propane as fuel, and the same extent of linearity was again observed over the same range of values for R_H and ϕ_F. From these results the linearity coefficients $k(\phi_F)$ can be determined for each fuel and are shown in Fig. 10a. It is seen that they behave similarly in terms of the variations with ϕ_F, with methane having higher values than ethene, propane, and butane. Furthermore, while it may be reasonable that propane and butane behave similarly as their controlling chemistries are quite similar, the fact that ethene, an alkene, has the same behavior as that of the alkanes is interesting because its chemistry is distinctively different and because it has much higher flame speeds.

In an effort to generalize the present result, and recognizing that Eq. (6) can be expressed in nondimensional and normalized form as:

$$\frac{s_u^o(\phi_F, R_H)}{s_u^o(\phi_F, 0)} = 1 + \left[\frac{k(\phi_F)}{s_u^o(\phi_F, 0)}\right] R_H \tag{9}$$

Figure 10b plots the nondimensional coefficient $K(\phi_F) = k(\phi_F)/s_u^o(\phi_F, 0)$ as a function of ϕ_F for the four fuels shown in Fig. 10a. If we interpret $K(\phi_F)$ as a normalized sensitivity coefficient characterizing the effect of hydrogen addition on the flame speed of a given fuel, then the results of Fig. 10b are quite reasonable in that methane, having the slowest flame speeds among the four fuels, would be most sensitive to the enhancement effect of hydrogen addition, and therefore has the largest values of $K(\phi_F)$. The opposite holds for ethene because it has the fastest flame speeds.

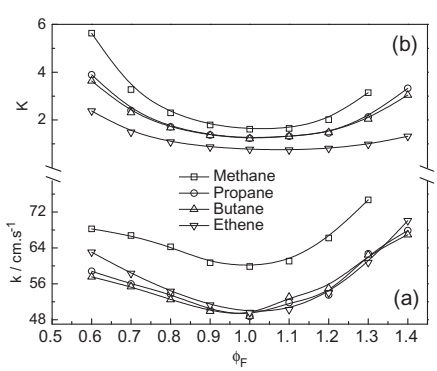

Fig. 10. Linearity coefficient (k) and normalized linearity coefficient (K) quantifying the increase in flame speed with R_H, for methane, ethene, propane, and butane.

6. Concluding remarks

In the present study, we have first experimentally acquired and then processed the data of high fidelity on the laminar flame speeds of butane–air mixtures with hydrogen addition, and showed that these results agree well with the previous finding in that the hydrogen-augmented flame speeds scale almost linearly with a suitably-defined hydrogen addition parameter, R_H. We then investigated the causes for the augmented burning rate, recognizing that hydrogen addition can modify the burning intensity through facilitated diffusion, intensified reactivity, and increased flame temperature, whose effects can be quantified through the effective Lewis number, global activation energy, and adiabatic flame temperature. Through a sensitivity-based analysis we have demonstrated that the kinetic effect is the most important, followed by the thermal effect, while the diffusion effect is minimal.

By further showing that the same behavior holds for methane, ethene, and propane, there exists the possibility of generalizing the present phenomena. As such, further study is merited.

Acknowledgments

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References

[1] B.E. Milton, J.C. Keck, *Combust. Flame* 58 (1) (1984) 13–22.
[2] S. Refael, E. Sher, *Combust. Flame* 78 (3–4) (1989) 326–338.
[3] Z. Huang, Y. Zhang, K. Zeng, B. Liu, et al., *Combust. Flame* 146 (1–2) (2006) 302–311.
[4] C. Tang, J. He, Z. Huang, C. Jin, J. Wang, et al., *Int. J. Hydrogen Energy* 33 (23) (2008) 7274–7285.
[5] G. Yu, C.K. Law, C.K. Wu, *Combust. Flame* 63 (3) (1986) 339–347.
[6] A.P. Kelley, C.K. Law, *Combust. Flame* 156 (9) (2009) 1844–1851.

[7] H. Wang, X. You, A.V. Joshi, S.G. Davis, A. Laskin, F. Egolfopoulos, C.K. Law, available at http://ignis.usc.edu/USC_Mech_II.htm.
[8] S.D. Tse, D.L. Zhu, C.K. Law, *Rev. Sci. Instrum.* 75 (1) (2004) 233–239.
[9] Z. Chen, M.P. Burke, Y.G. Ju, *Proc. Combust. Inst.* 32 (2009) 1253–1260.
[10] A.P. Kelley, G. Jomaas, C.K. Law, *Combust. Flame* 156 (5) (2009) 1006–1013.
[11] M.P. Burke, Z. Chen, Y.G. Ju, F.L. Dryer, *Combust. Flame* 156 (4) (2009) 771–779.
[12] K.J. Bosschaart, L.P.H. de Goey, J.M. Burgers, *Combust. Flame* 136 (3) (2004) 261–269.
[13] S.G. Davis, C.K. Law, *Combust. Sci. Technol.* 140 (1) (1998) 427–449.
[14] C.K. Law, C.J. Sung, *Prog. Energy Combust. Sci.* 26 (4–6) (2000) 459–505.
[15] C.K. Law, *Combustion Physics*, Cambridge University Press, New York, USA, 2006, p. 281.
[16] G. Jomaas, C.K. Law, J.K. Bechtold, *J. Fluid Mech.* 583 (2007) 1–26.
[17] C.K. Law, G. Jomaas, J.K. Bechtold, *Proc. Combust. Inst.* 30 (1) (2005) 159–167.
[18] J.K. Bechtold, M. Matalon, *Combust. Flame* 127 (1–2) (2001) 1906–1913.